

Chapter 8

Soil Resources

Class 10 - Total Geography Morning Star

Choose the correct option

Question 1

Which of the following constituents add fertility to the soil?

1. Silica
2. Humus
3. Clay
4. Sand

Answer

Humus

Question 2

Which of the following are the characteristics of fertile soil?

1. Adequate amount of moisture
2. Rich in nutrients
3. Organic matter
4. All of the above.

Answer

All of the above

Question 3

Which of the following is a transported or ex situ soil?

1. Alluvial soil
2. Laterite soil

3. Red soil
4. Black soil.

Answer

Alluvial soil

Question 4

Which of the following soils vary from place to place on the basis of the parental rocks and climatic conditions?

1. Black soil
2. Laterite soil
3. Alluvial soil
4. Red soil.

Answer

Red soil

Question 5

Farmers plant trees in several rows to check soil erosion by

1. floods
2. winds
3. drought
4. heavy rains

Answer

winds

Question 6

Alluvial soil is rich in which of the following minerals?

1. Iron
2. Nitrogen
3. Phosphorus
4. None of the above.

Answer

Iron

Question 7

Red soil gets its name from

1. its colour
2. its ferric oxide content
3. Both (a) and (b)
4. None of the above.

Answer

Both (a) and (b)

Question 8

Black soil is also known as

1. Bhangar
2. Khadar
3. Regur
4. Leached

Answer

Regur

Question 9

Laterite soil is mainly found in the highland areas of

1. the Himalayas
2. the Peninsular Plateau
3. the Aravali Range
4. the Karakoram Range

Answer

the Peninsular Plateau

Question 10

Why is Black soil highly moisture retentive?

1. It has fifty per cent of clay content
2. It is formed of weathered lava rocks
3. It has high quantities of lime and iron
4. All of the above.

Answer

It has fifty per cent of clay content

Question 11

Which of the following soils forms the second largest soil group in India?

1. Black soil
2. Red soil
3. Yellow soil
4. Brown soil

Answer

Red soil

Question 12

Why is the Red soil considered ideal for dry farming?

1. It is formed in areas with poor rainfall and responds to fertilizers.
2. It is red in colour.
3. It is rich in iron oxide.
4. It is rich in nitrogen.

Answer

It is formed in areas with poor rainfall and responds to fertilizers.

Question 13

Which of the following is correct about the Laterite soil?

1. It is the residual soil.
2. It is formed by leaching due to tropical rains.
3. Humus content of the soil is removed by bacteria.
4. All of the above.

Answer

All of the above

Answer the following questions

Question 1(a)

What is the soil? Name the four types of soil.

Answer

Soil is the thin mixture of small rock particles and rotting organic matter that covers the surface of the earth. The four types of soil are-

1. Alluvial soil
2. Black soil
3. Red soil
4. Laterite soil

Question 1(b)

State any two factors responsible for the formation of soil.

Answer

Two factors responsible for the formation of soil are —

1. Weathering
2. Erosion

Question 1(c)

Give a geographical reason for each of the following:

- (i) The colour of alluvium soil varies in different areas.
- (ii) Khadar is more fertile than bhangar soil.
- (iii) Black soil has a kind of self-ploughing characteristic.

Answer

(i) The colour of alluvium soil varies in different areas because its shades depend on the depth of deposition, the texture of the materials and the time taken for maturity.

(ii) Khadar is more fertile than bhangar soil as new layers are deposited year after year during monsoon floods.

(iii) Black soil has a kind of self-ploughing characteristic because during dry season, it shrinks and develops big cracks which help in air circulation. This characteristic causes a kind of self-ploughing.

Question 1(d)

(i) How can you say that soil has evolved over millions of years?

(ii) Name one important constituent of soil that adds fertility to the soil.

Answer

(i) Soil changes with age - As a soil ages, it gradually starts to look different from its parent material. That is because soil is dynamic. Its components—minerals, water, air, organic matter, and organisms—constantly change.

(ii) One important constituent of soil that adds fertility to the soil is humus.

Question 2(a)

How are alluvial soils formed?

Answer

Alluvial soil is formed by the sediments brought down by rivers. The rivers deposit very fine particles of soil called alluvium in their plains during the course of their long journey. The resultant alluvial soil is known as riverine soil.

Question 2(b)

Mention any two advantages of alluvial soil.

Answer

Two advantages of alluvial soil are-

1. It is light and porous. So, it is easily tillable.
2. It is suitable for the growth of a large variety of rabi and kharif crops.

Question 2(c)

Give a geographical reason for each of the following:

(i) Different regions in India have different kinds of soil for agriculture.

(ii) Alluvial soils vary in texture.

(iii) Nearly all types of crops grow well in riverine soils.

Answer

(i) Different regions in India have different kinds of soil for agriculture because India is a vast region with varied natural environment. Several physiographic regions and different types of climate lead to the formation of different types of soil in different regions.

(ii) Alluvial soils vary in texture as in the upper reaches of the river valleys, the soil is dry, coarse, porous and sandy. Soil particles are large and non-uniform but as we move further down a river valley, the soil particles become smaller and more uniform. They are more compact and more moist.

(iii) Nearly all types of crops grow well in riverine soils because alluvial soil is a mixture of sand, clay and silt. It is rich in mineral and easily tillable. Its composition makes it suitable for the growth of a variety of crops.

Question 2(d)

(i) Why is soil a valuable resource for India?

(ii) Why is black soil considered a productive soil?

Answer

(i) Soil is a valuable resource for India because agriculture is the main occupation in India. Soil supports agriculture and it is the source of nutrients for many forms of life.

(ii) Black soil is considered a productive soil because it has high moisture retention capacity and more clay content which responds well to irrigation. It is suitable for the cultivation of cotton, jowar, wheat, linseed, gram, citrus fruits and vegetables, tobacco and sugarcane.

Question 3(a)

Give two characteristic features of the soil found most suitable for growing cotton and sugarcane in Maharashtra.

Answer

Black soil is suitable for growing cotton and sugarcane in Maharashtra. Its characteristics are-

1. It is black in colour.
2. It has high quantities of lime, iron, magnesium and generally poor percentage of phosphorus, nitrogen and organic matter.

Question 3(b)

Name one soil of volcanic origin commonly found in India. Name one crop widely grown on this soil.

Answer

One soil of volcanic origin commonly found in India is black soil. Cotton is widely grown on this soil.

Question 3(c)

Give a geographical reason for each of the following:

- (i) Deltaic alluvium is more fertile than the coastal alluvium.
- (ii) Black soil is largely found in the Deccan Trap region.
- (iii) Black soil is often referred to as Cotton soil or Regur soil.

Answer

(i) Deltaic alluvium is more fertile than coastal alluvium as new layers are deposited year after year during monsoonal floods. Also, deltaic alluvium is composed of fine silt and clay whereas coastal alluvium is full of lime nodules.

(ii) Black soil is found in the Deccan trap region because this region is formed due to the deposition of lava produced by the volcanic eruption. When lava rocks break down into finer particles due to weathering, it results in the formation of the black soil.

(iii) Black soil is often referred to as Cotton soil or Regur soil because it is black in colour and is suitable for cotton cultivation.

Question 3(d)

- (i) Name the process by which Laterite soil is formed.
- (ii) Mention one disadvantage of Laterite soil.

Answer

(i) Laterite soil is formed as a result of atmospheric weathering of rocks under the conditions of high rainfall and temperature with alternate wet and dry periods. It is the residual soil formed by leaching due to tropical rains.

(ii) One disadvantage of Laterite soil is that it does not retain moisture and hence, is infertile.

Question 4(a)

State the two sub-categories into which alluvial soil is generally divided. Which one of them is superior to the other? Why?

Answer

Two sub-categories into which alluvial soil is generally divided are khadar and bhangar.

Khadar is superior to bhangar because it is more fertile as new layers are deposited year after year during monsoon floods.

Question 4(b)

What are the characteristics of alluvial soil?

Answer

The characteristics of alluvial soil are:

1. **Colour** — The colour of alluvial soil varies from light grey to deep black.
2. **Texture** — In the upper reaches of the valleys, the soil is dry, coarse, porous and sandy with large and non-uniform soil particles. In the lower reaches of the river valley, the soil particles become smaller and more uniform and more moist.
3. **Minerals** — It varies in nature from sandy loam to clay. It is fertile with minerals like iron, magnesia, alumina, potash and lime but poor in phosphorus, nitrogen and organic matter.

Question 4(c)

Give a geographical reason for each of the following:

- (i) Laterite soil is useful for construction purpose.
- (ii) Soils of the Western Ghats are affected by wind erosion.
- (iii) The foothills of the Himalayas are prone to excessive soil erosion.

Answer

(i) Laterite soils are used as a building material because they become hard like iron when exposed to air. So, they are used as a building material. Laterite soils are red in color due to the presence of iron oxide. These soils are generally coarse in texture and friable and porous in nature.

(ii) Soils of the Western Ghats are affected by wind erosion because a large chunk of Western zone is arid or semi-arid and wind erosion is a result of extreme form of soil degradation due to aridity.

(iii) The foothills of the Himalayas are prone to excessive soil erosion because the vegetation over there is very less. There are not many plant or tree roots to hold the soil in place, which makes the foothills of the Himalayas prone to excessive soil erosion.

Question 4(d)

(i) Which soil is suitable for growing coffee in Karnataka?

(ii) Mention two main characteristics of this soil.

Answer

(i) Laterite soil is suitable for growing coffee in Karnataka.

(ii) Two main characteristics of this soil are-

1. It is red in colour due to the presence of iron oxide.
2. It is of a coarse texture, soft and friable.

Question 5(a)

Name two States in India where Regur soil is found.

Answer

Regur soil is found in Maharashtra and Madhya Pradesh.

Question 5(b)

Give two points of difference between Regur and Alluvial soil.

Answer

Regur soil	Alluvial soil
It is formed by the weathering of lava rocks.	It is formed by deposition of sediments by river.
It is of black colour.	Its colour varies from light grey to deep black.

Question 5(c)

Give a geographical reason for each of the following:

(i) Jowar is grown in Maharashtra.

(ii) Bajra is grown in Rajasthan.

(iii) Sugarcane is grown in Uttar Pradesh.

Answer

(i) Jowar is grown in Maharashtra because of the black regur soil which is rich in lime, potash-magnesium, carbonates, allumina and calcium.

(ii) Bajra is grown in Rajasthan because of the sandy soil found in this region. It is rich in soluble salts.

(iii) Sugarcane is grown in Uttar Pradesh because the climatic conditions and presence of alluvial soil support the cultivation of sugarcane.

Question 5(d)

(i) Besides cotton, name the crops which grow well in Regur soil.

(ii) In what way does Regur soil help agriculture?

Answer

(i) Besides cotton, jowar, wheat, linseed, gram, citrus fruits and vegetables, tobacco and sugarcane grow well in Regur soil.

(ii) Regur soil helps agriculture as it is very fertile. In any season, it has moisture stored in its subsoil. This allows dry farming as well. In dry season, it forms big cracks which help in air circulation, and causes a kind of self ploughing.

Question 6(a)

Name two states where red soil is found. Mention two important crops grown in this soil.

Answer

Two states where red soil is found are Tamil Nadu and Karnataka. Two important crops grown in this soil are rice and tobacco.

Question 6(b)

State two advantages of this type of soil.

Answer

Two advantages of this type of soil are-

1. It is porous, loose and aerated.

2. It is ideal for dry farming.

Question 6(c)

Give a geographical reason for each of the following:

- (i) Black soil is also called lava soil.
- (ii) Red soil is red in colour and its colour varies from yellow to brown.
- (iii) Jute is cultivated in West Bengal.

Answer

- (i) Black soil is also called lava soil because it is formed by the denudation of volcanic rocks.
- (ii) Red soil is red in colour as it contains large amounts of iron oxide. At several places, its colour changes and it appears brown or grey. It looks yellow when it occurs in a hydrated form.
- (iii) Jute is cultivated in West Bengal because the alluvial soil found in West Bengal is very fertile and supports jute cultivation.

Question 6(d)

- (i) How is red soil formed?
- (ii) How is this soil suitable for dry farming?

Answer

- (i) Red soil develops on old crystalline rocks. Under prolonged weathering by rainfall, ancient crystalline and metamorphic rocks of the peninsular plateau break up to form this soil.
- (ii) This soil is suitable for dry farming as it is formed in areas with poor rainfall.

Question 7(a)

What is soil erosion? Name the areas of soil erosion in India-

- (i) for wind erosion
- (ii) for running water erosion

Answer

The removal of the top soil cover by water, wind and human activities is called soil erosion.

- (i) Wind erosion affects the dry areas of Rajasthan, Gujarat and Haryana

(ii) Running water erosion affects the badlands of Chambal and Yamuna river, the piedmont zone of western Himalayas, the Chotanagpur plateau, the Tapi-Sabarmati valley and the regur soil area of Maharashtra.

Question 7(b)

Give two methods used to prevent soil erosion.

Answer

Two methods used to prevent soil erosion are-

1. Terrace farming
2. Shelter belts

Question 7(c)

Give a geographical reason for each of the following:

- (i) Shelter belts reduce soil erosion in arid areas.
- (ii) Rural development will influence soil conservation.
- (iii) Water harvesting aids in soil conservation.

Answer

(i) Shelter belts reduce soil erosion in arid areas because planting trees in several rows checks wind erosion.

(ii) Rural development will influence soil conservation as in the programmes of rural development, villagers are provided with the techniques of how to improve vegetative cover and the steps required to be taken for preventing soil erosion. The farmers are briefed about the farming methods (such as strip cropping and shelter belts) to be followed to reduce soil erosion.

(iii) Water harvesting aids in soil conservation as it reduces soil erosion and flood hazards by collecting rainwater and reducing the flow of water which causes the removal of fertile top soil.

Question 7(d)

- (i) What is meant by soil conservation?
- (ii) State two measures to conserve soil in hilly areas.

Answer

(i) Soil conservation means prevention of soil loss from erosion or prevention of reduced fertility of soil caused by over-use, acidification, salinisation or other types of soil contamination.

(ii) Two measures to conserve soil in hilly areas are-

1. Terrace farming- Terraces act as bunds and prevent soil erosion by running water.

2. Contour Ploughing- Ploughing along contours on a slope prevents soil erosion by running water.

Question 8(a)

How does deforestation lead to soil erosion?

Answer

Deforestation exposes the soil to water and wind, which leads to soil erosion.

Question 8(b)

State two methods of controlling erosion of soil caused by running water.

Answer

Two methods of controlling erosion of soil caused by running water are construction of dams and plugging gullies.

Question 8(c)

Name two important agents of erosion. For each, state one method of controlling the erosion caused.

Answer

Two important agents of erosion are water and wind. Water erosion can be controlled by the construction of dams. Wind erosion can be controlled by making shelter belts.

Question 8(d)

Explain briefly the need for conservation of soil as a natural resource.

Answer

There is an acute need for conservation of soil as a natural resource because of its adverse effects which include the following-

1. Loss of fertile topsoil with its mineral content

2. Lowering of the water table
3. Drying of vegetation and extension of arid lands leading to an increase in the frequency of drought and floods
4. Silting of river and canal beds
5. Recurrence of landslides
6. Adverse effect on economic prosperity and cultural development

Thinking Skills

Question 1

Suppose you are given two bags of soil, one containing bhangar soil and the other containing khadar soil. But nothing is written on the bags about the soil types. How will you distinguish the two soil types?

Answer

On the basis of its age, the alluvial soil is of two types — bhangar and khadar. Both of them have different appearance and texture.

The bhangar soil or the older alluvium is composed of lime nodules or kankars and has a clayey composition. The khadar soil is light in colour and is composed of newer deposits. It is formed of fine silt and clay.

So by examining the appearance and texture of the two soils, I will distinguish between the bag of khadar and bhangar soil.

Question 2

A farmer bought land in an area having poor rainfall, with soil which is suitable for dry farming. Name the type of soil available in this area. Which crops will he be able to grow there?

Answer

Red soil is available in this area as this type of soil is formed in areas with poor rainfall.

Though red soil is suitable for cultivating almost all crop types, it is most suitable for growing vegetables, rice, ragi and tobacco. Groundnut and potatoes can be grown on coarse soils at higher levels and sugarcane on heavy clays at lower levels.

Question 3

Your uncle lives in a hilly area which gets heavy rainfall. He grows certain special crops like tea, coffee, tapioca and cashewnuts. Besides agriculture, he has a brick kiln as the area has ample raw

material for making bricks. Name the area where your uncle may be residing. What type of soil is available in this area and how is it formed?

Answer

My uncle is residing in the highland areas of the Peninsular Plateau.

Laterite soil is available in this area.

Laterite soil is formed as a result of atmospheric weathering of rocks under conditions of high rainfall and temperature with alternate wet and dry periods. It is the residual soil formed by leaching due to tropical rains. Leaching or desilication is the process in which the nutrients get percolated down below the soil due to heavy rainfall; thus leaving the topsoil infertile. Due to heavy rains, lime and silica are leached away and aluminium compounds are left behind. Humus content of the soil is removed by bacteria that survives well in high temperature.

Question 4

You had gone for an excursion to a mountainous region along with your teachers and classmates. On reaching there you were shocked to see bare mountains with no vegetation or greenery. What reasons did your teacher give for such a deplorable conditions of the mountains? List the solutions to rectify the problem.

Answer

I had gone for an excursion to a mountainous region with my teachers and classmates. I was shocked to see bare mountains with no vegetation or greenery. When I asked the reason for such a condition of the mountain, my teacher told me that the main reason was soil erosion. There are various causes of soil erosion:

1. The loss of vegetation cover leads to Sheet Erosion on hilly slopes because water instead of sinking into the ground washes the soil down.
2. The area also receives heavy monsoon rains which causes Sheet, Rill and Gully erosion.
3. In this area, local population practise shifting cultivation. The heavy rains then wash away the bare soil from the slopes to the valleys below.
4. Another important cause of Sheet, Rill and Gully Erosion is uncontrolled grazing of domestic animals in the upper slopes.

The solutions to rectify the problem of soil erosion are as follows:

1. **Terrace Farming** — On hilly slopes, terraces act as bunds and prevent the soil from being washed away by running water.
2. **Contour Ploughing** — Ploughing along contours on a slope prevents soil being washed away by rainwater or by surface run off. Contours act like bunds. Terraces are levelled into step-like small fields with even slope.

3. **Plugging Gullies** — The gullies made in the soil are plugged with deposition of silt during heavy rains.
4. **Planting Trees** — The trees are planted along the edges of the fields, the waste land and on steep slopes to prevent soil erosion as well as to enhance the capacity of the soil to retain water.